

CHELTENHAM GIRLS 2007

EXN II

Question 1 (15 marks) Start a NEW page

- (a) Let $\alpha = 3 + 4i$ and $\beta = 1 - i$
Express in the form $x + iy$ where x and y are real
- (i) $\alpha\beta$ 1
- (ii) $\frac{\alpha}{\beta}$ 2
- (iii) $(\overline{\beta})^2$ 2
- (b) Consider the equation $z^2 + \gamma z + (2 - i) = 0$
Find the complex number γ , given that $2i$ is a root of the given equation. 2
- (c) Let $\beta = -1 - i\sqrt{3}$
- (i) Express β in modulus-argument form. 2
- (ii) Express β^{-10} in modulus-argument form. 2
- (iii) Hence express β^{-10} in the form $x + iy$ 1
- (d) Shade the region in the number plane described by
 $\frac{\pi}{3} < \arg z < \pi$ and $1 \leq |z| \leq 3$ 3

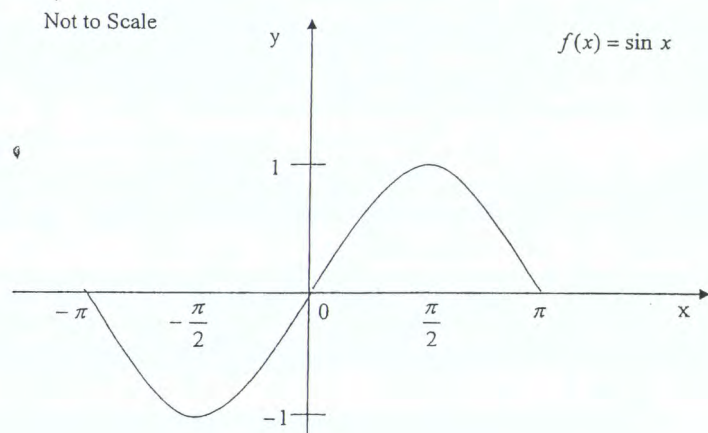
Question 2 (15 marks) Start a NEW page

- (a) (i) Find real numbers a and b such that
$$\frac{1}{(3-x)(1+x)} \equiv \frac{a}{(3-x)} + \frac{b}{(1+x)}$$
 2
- (ii) Hence find $\int \frac{1}{(3-x)(1+x)} dx$ 2
- (b) Find $\int \frac{1}{e^x + e^{-x}} dx$ 3
- (c) By completing the square, find
$$\int \frac{3}{\sqrt{x^2 + 6x + 13}} dx$$
 2
- (d) (i) Evaluate $\int_0^{\frac{\pi}{2}} \frac{1}{1 + \cos x + \sin x} dx$,
using the substitution $t = \tan \frac{x}{2}$ 3
- (ii) Hence evaluate $\int_0^{\frac{\pi}{2}} \frac{x}{1 + \cos x + \sin x} dx$,
using the substitution $u = \frac{\pi}{2} - x$ 3

42

Question 3 (15 marks) Start a NEW page

- (a) The diagram shows the graph of $f(x) = \sin x$, for $-\pi \leq x \leq \pi$



- (b) (i) Sketch on the same number plane: 2

$$y = |x| - 2 \text{ and}$$

$$y = -x^2 + 3x$$

- (ii) Hence or otherwise solve $\frac{|x| - 2}{-x^2 + 3x} > 0$ 2

- (c) α, β, γ are the roots of the equation $x^3 + bx^2 + 12x + 4 = 0$, where b is a real constant.

- (i) Find an equation with $\alpha^{-1}, \beta^{-1}, \gamma^{-1}$ as roots. 2

- (ii) Hence, or otherwise, find the value of b , given that 2

$\alpha^{-1}, \beta^{-1}, \gamma^{-1}$ are in arithmetic progression.

Draw separate one-third page sketches of the graphs of the following:

(i) $y = \frac{1}{f(x)}$ 1

(ii) $y = (f(x))^2$ 2

(iii) $|y| = f(x)$ 2

(iv) $y = \sin^{-1} f(x)$ 2

Question 3 continues on page 4

End of Question 3

43

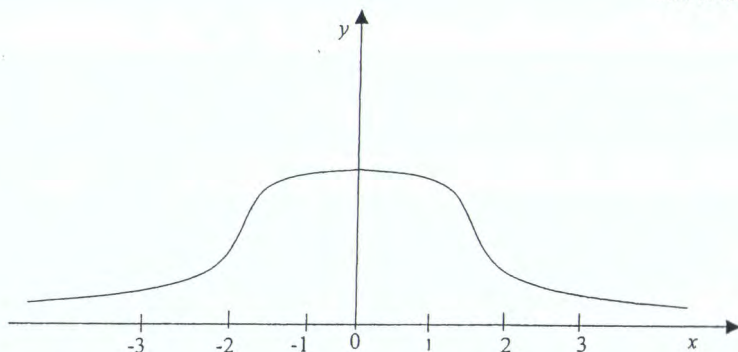
Question 4 (15 marks) Start a NEW page

- (a) The area bounded by the curve $y = e^{-\frac{1}{2}x^2}$ and the x -axis, between $x = 0$ and $x = 2$ is rotated about the y -axis. Using the method of cylindrical shells, find the volume of the solid formed. (Leave your answer in exact form)

3

$$y = e^{-\frac{1}{2}x^2}$$

Not to Scale



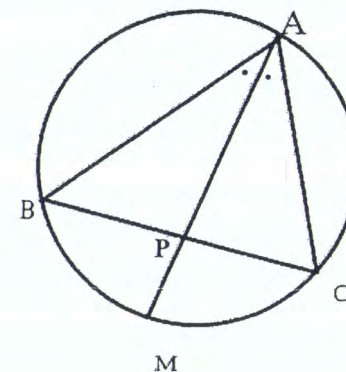
- (b) Consider the conic defined by the equation $\frac{x^2}{19-l} + \frac{y^2}{7-l} = 1$

- (i) Determine the real value of l for which the equation defines: 2
 (α) an ellipse
 (β) an hyperbola
- (ii) Sketch the curve corresponding to $l = 3$. Show the foci and the x and y intercepts. You are NOT required to show the directrices. 2
- (iii) Describe how the shape of this curve changes as l increases from 3 towards 7. 1
- (iv) Describe the limiting shape of the curve as l approaches 7. 1

Question 4 continues on page 6

- (c) A circle is drawn to pass through the vertices of $\triangle ABC$. AM bisects $\angle BAC$ and meets BC at P as shown in the diagram.

Not to Scale



Copy or trace this diagram onto your answer sheet.

- (i) Prove that $\triangle ABM$ and $\triangle ACP$ are similar. 2
- (ii) Prove that $AB \cdot AC = AP \cdot AM$ 1
- (iii) Hence prove that $AB \cdot AC - BP \cdot PC = AP^2$ 3

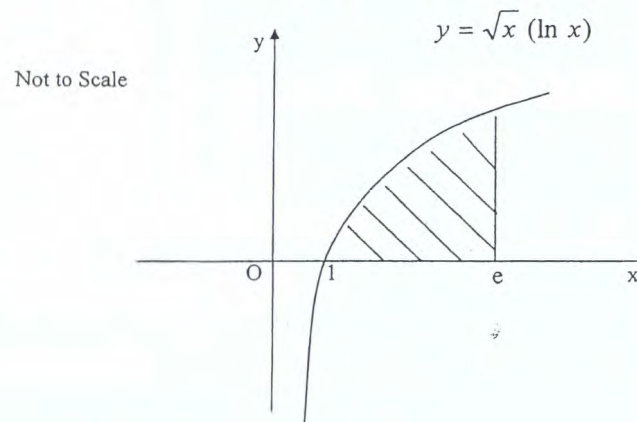
End of Question 4

Question 5 (15 marks) Start a NEW page

- (a) (i) Show that $2 \cos A \sin B = \sin(A + B) - \sin(A - B)$ 1
- (ii) Hence, or otherwise, show that
 $(1 + 2 \cos \theta + 2 \cos 2\theta + 2 \cos 3\theta) \sin \frac{\theta}{2} = \sin \frac{7\theta}{2}$ 2
- (iii) Let $A = 1 + 2 \cos \theta + 2 \cos 2\theta + 2 \cos 3\theta$ 1
 Show that if $\theta = \frac{2\pi}{7}$, then $A = 0$
- (iv) Express A in terms of $\cos \theta$ 1
 You may assume $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$. Do NOT prove this.
- (v) Hence or otherwise prove that $\cos \frac{2\pi}{7}$ 1
 is a solution of the polynomial equation $8x^3 + 4x^2 - 4x - 1 = 0$

Question 5 continues on page 8

- (b) (i) Let $I_n = \int_1^e x(\ln x)^n dx$, $n = 0, 1, 2, \dots$ 2
 Use integration by parts to show that
 $I_n = \frac{e^2}{2} - \frac{n}{2} I_{n-1}$ $n = 1, 2, 3, \dots$
- (ii) The area bounded by the curve $y = \sqrt{x} (\ln x)$, $x \geq 1$,
 the x -axis and the line $x = e$ is rotated about the x -axis.
 Find the exact volume of the solid of revolution so formed. 3



- (c) 20 teachers volunteer to be considered for a special 50th anniversary celebrations committee. These consist of 4 Maths teachers, 4 Art teachers, 6 English teachers and 6 History teachers. 9 teachers are to be randomly chosen to form the committee.
- (i) If there are no restrictions, in how many ways can the 9 teachers be chosen? 1
- (ii) In how many ways can the 9 teachers be chosen if no Art or Maths teachers are included? 1
- (iii) What is the probability that no more than 1 Maths teacher and no more than 1 Art teacher are on the committee? 2

End of Question 5

45

Question 6 (15 marks) Start a NEW page

- (a) A body of mass m is falling from rest and experiences air resistance of kv^2 per unit mass. k is a constant, g is gravity under acceleration and v is the velocity of the body.

(i) Show that the equation of motion of the body is given by 1

$$\ddot{x} = g - kv^2$$

(ii) If V is the terminal velocity, show that $V = \sqrt{\frac{g}{k}}$ 1

(iii) Show that $x = \frac{1}{2k} \ln \left(\frac{g}{g - kv^2} \right)$, where x is the distance travelled. 4

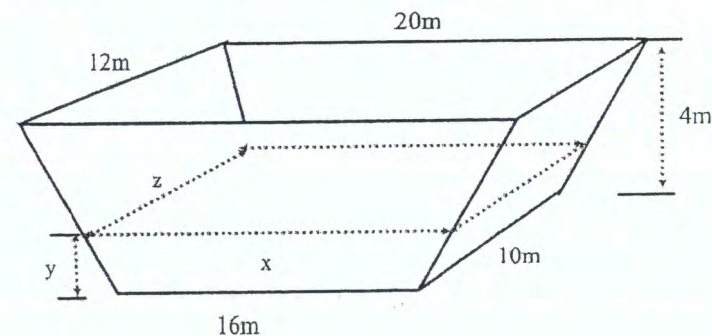
(iv) If W is the velocity of the body when it reaches the ground, show that the distance, S , fallen is given by 1

$$S = \frac{1}{2k} \ln \left(\frac{V^2}{V^2 - W^2} \right)$$

Question 6 continues on page 10

- (b) The Bundeena Voluntary Firefighters have a massive water storage tank. The tank has a rectangular base with sides 16 metres and 10 metres. Its top is also rectangular with dimensions 20 metres and 12 metres. The tank has a depth of 4 metres and each of its four side faces is an isosceles trapezium. Each horizontal cross section parallel to the base of the tank is a rectangle.

Not to Scale



- (i) Consider a cross section of the tank x metres by z metres and height y above the base. Show that the area of this cross section is given by 3

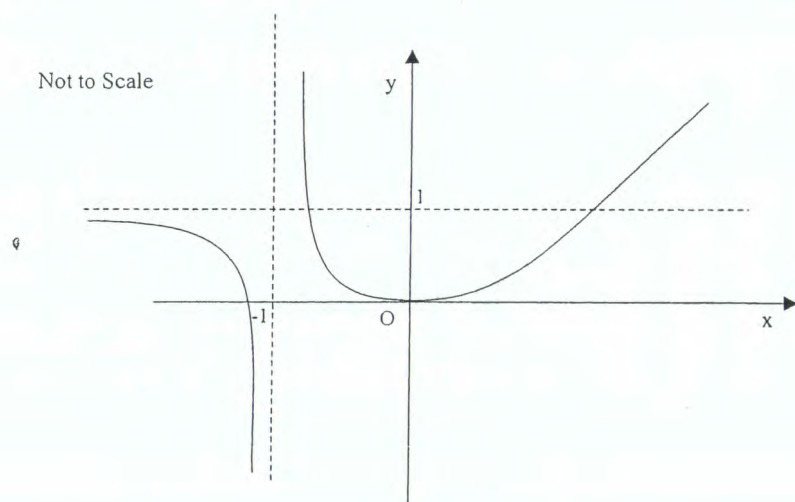
$$\frac{1}{2}y^2 + 18y + 160$$

- (ii) Hence find the volume of the tank. 2

Question 6 continues on page 11

(c)

3



The diagram shows the graph of $y = f'(x)$, the gradient function of $y = f(x)$

Copy or trace this diagram onto your answer sheet.

Sketch, on a separate number plane and underneath your copy, the graph of $y = f(x)$, given that $f(0) = 1$ and $f(x) < 0$ for $x < -1$. Show all relevant asymptotes.

End of Question 6

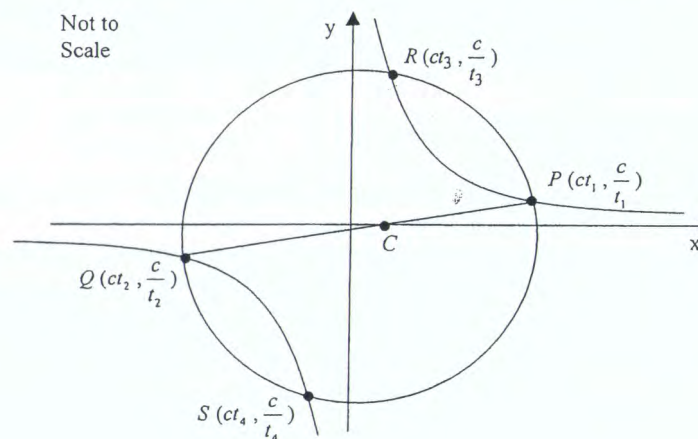
Question 7 (15 marks) Start a NEW page

- (a) Let P , Q , R and S be four points with parameters t_1 , t_2 , t_3 and t_4 on the hyperbola $x = ct$, $y = \frac{c}{t}$, as shown in the diagram.

The hyperbola is intersected by a circle whose equation is

$$(x - g)^2 + y^2 = r^2$$

P , Q , R and S lie on the circle, as shown and C is the centre of the circle.

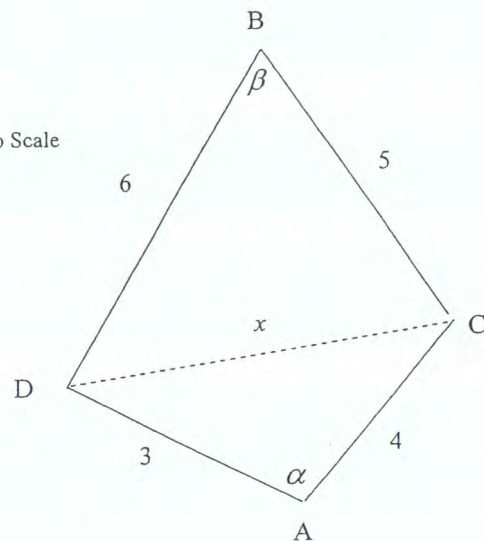


- (i) Show that $t_1 + t_2 + t_3 + t_4 = \frac{2g}{c}$ 2
- (ii) If C is the midpoint of PQ , show that $t_1 + t_2 = \frac{2g}{c}$ 2
- (iii) Hence, or otherwise, show that the origin is the midpoint of RS . 2

Question 7 continues on page 13

(b)

Not to Scale



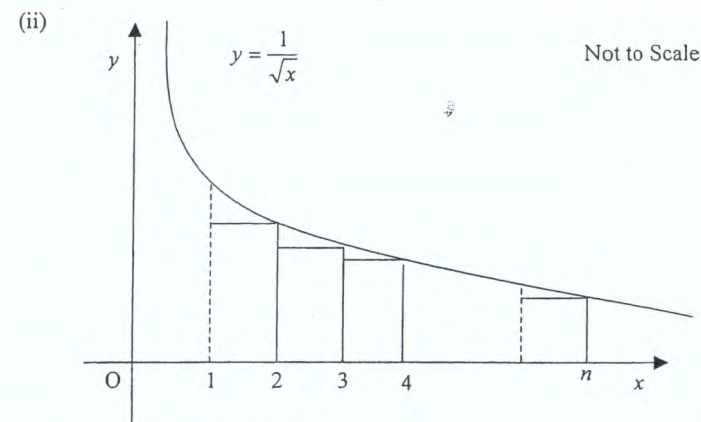
The quadrilateral $ACBD$ has sides $AC = 4$, $CB = 5$, $BD = 6$ and $DA = 3$.
 $\angle DAC = \alpha$ and $\angle DBC = \beta$. $DC = x$, as shown.

- | | | |
|-------|--|---|
| (i) | Show that the area of $ACBD$ is given by
$A = 6 \sin \alpha + 15 \sin \beta$ | 1 |
| (ii) | By equating expressions for x^2 , show that
$2 \cos \alpha - 5 \cos \beta = -3$ | 2 |
| (iii) | Differentiate the identity in part (ii) implicitly,
with respect to α ,
to find an expression for $\frac{d\beta}{d\alpha}$. | 1 |
| (iv) | Prove that the area of the quadrilateral is a maximum when
the quadrilateral is cyclic. You need NOT prove the
relevant stationary point is a maximum. | 3 |
| (v) | If the quadrilateral is cyclic, find the exact area. | 2 |

End of Question 7

Question 8 (15 marks) Start a NEW page

- | | | | |
|-----|------|---|---|
| (a) | (i) | Find a general solution for the equation
$\cos A = \cos B$
Write the answer in terms of A and B . | 1 |
| | (ii) | Hence, or otherwise, solve
$\cos 2\theta = \sin 3\theta$, for $0 \leq \theta \leq \pi$ | 3 |
| (b) | (i) | Use Mathematical Induction to show that
$\sum_{r=1}^n \frac{1}{\sqrt{r}} > 2(\sqrt{n+1} - 1)$, where $n = 1, 2, 3, \dots$
You may assume the inequality:
$2k+3 > 2\sqrt{(k+1)(k+2)}$ Do NOT prove this. | 3 |



Use the graph of $y = \frac{1}{\sqrt{x}}$ to show that

$$\sum_{r=1}^n \frac{1}{\sqrt{r}} < 1 + \int_1^n \frac{1}{\sqrt{x}} dx$$

- | | | |
|-------|---|---|
| (iii) | Hence show that | 3 |
| | $198 < \sum_{r=1}^{10000} \frac{1}{\sqrt{r}} < 199$ | |

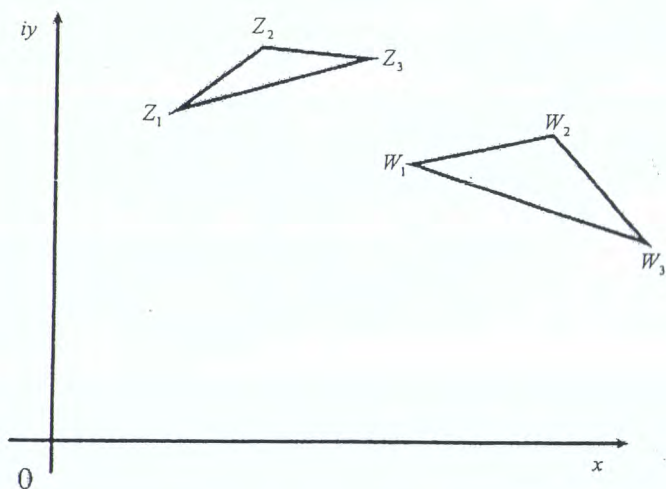
Question 8 continues on page 15

48

- (c) The points Z_1, Z_2, Z_3 represent the complex numbers z_1, z_2, z_3 and the points W_1, W_2, W_3 represent the complex numbers w_1, w_2, w_3 .

If $\frac{z_2 - z_1}{z_3 - z_1} = \frac{w_2 - w_1}{w_3 - w_1}$ prove that $\triangle Z_1 Z_2 Z_3$ and $\triangle W_1 W_2 W_3$ are similar.

Not to Scale



End of Paper

① $z = 3 + 4i$, $\beta = 1 - i$

(i) $\alpha\beta = (3 + 4i)(1 - i)$
 $= 3 - 3i + 4i + 4$
 $= 7 + i$

(ii) $\frac{z}{\beta} = \frac{3 + 4i}{1 - i} \times \frac{1 + i}{1 + i}$
 $= \frac{3 + 3i + 4i - 4}{1 + 1}$
 $= \frac{-1 + 7i}{2}$
 $= -\frac{1}{2} + \frac{7}{2}i$

(iii) $(\frac{z}{\beta})^2 = (1 + i)^2$
 $= 1 + 2i - 1$
 $= 2i$

Solutions.

①

②

②

(b) $z + v + w + (2 - i) = 0$

Sub $z = 2i$

$(2i) + 8 + 2i + 2 - i = 0$

$-4 + 2i + 2 - i = 0$

$-2 + 2i - i = 0$

$\times$ by i

$-2i - 2 + 1 = 0$

$+2i = 1 - 2i$

$2(x + iy) = 1 - 2i$

$2x = 1$, $2y = -2$

$x = \frac{1}{2}$, $y = -1$

$z = \frac{1}{2} - i$

(c) $\beta = -1 - i\sqrt{3}$

(i) $|\beta| = \sqrt{1 + 3}$

$= 2$

$\tan \theta = \frac{1}{\sqrt{3}}$

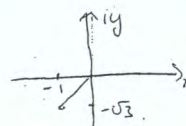
$\Rightarrow \theta = \frac{\pi}{3}$

$\arg \beta = -\frac{2\pi}{3}$

$\beta = 2 \operatorname{cis}(-\frac{2\pi}{3})$

(ii) $\beta^{-10} = 2^{-10} \operatorname{cis}(\frac{20\pi}{3})$
 $= 2^{-10} \operatorname{cis}(\frac{2\pi}{3})$

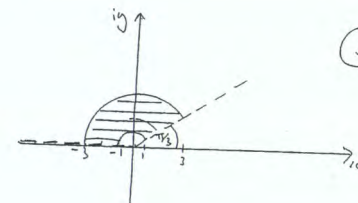
(iii) $\beta^{-10} = 2^{-10} (\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3})$
 $= 2^{-10} (\frac{-1}{2} + i \frac{\sqrt{3}}{2})$
 $= -2^{-11} + i(\sqrt{3})2^{-11}$



②

① (d)

$\arg z > \frac{\pi}{3}$, $1 \in \operatorname{Re} z \leq 3$



③

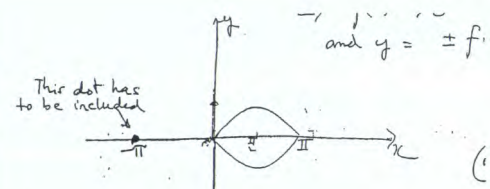
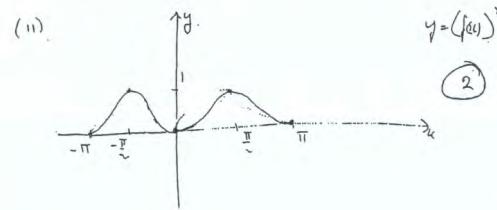
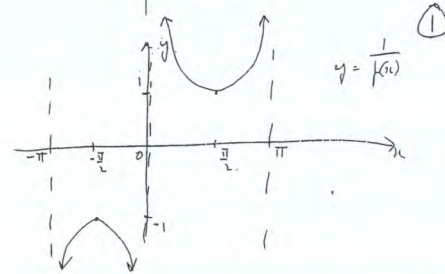
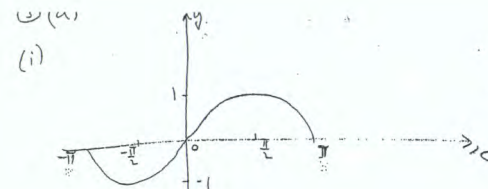
49

p2

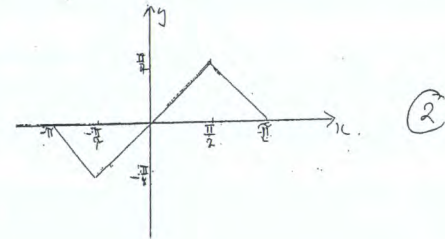
(a) (i) $\frac{1}{(3-x)(1+x)} = \frac{a}{3-x} + \frac{b}{1+x}$
 $= \frac{a(1+x) + b(3-x)}{(3-x)(1+x)}$
 $= \frac{a(1+x) + b(3-x)}{(3-x)(1+x)}$
 $\text{Let } x=1: 1 = \frac{4b}{4} \Rightarrow b=1$
 $\text{Let } x=3: 1 = \frac{4a}{4} \Rightarrow a=1$

(ii) $\int \frac{1}{(3-x)(1+x)} dx = \frac{1}{4} \int \frac{1}{3-x} + \frac{1}{1+x} dx$
 $= \frac{1}{4} \ln|3-x| + \frac{1}{4} \ln|1+x| + C$
 $= \frac{1}{4} \ln \left| \frac{1+x}{3-x} \right| + C$

(b) $\int \frac{e^x + e^{-x}}{e^x + 1} dx$
 $= \int \frac{1}{e^x + 1} dx$
 $= \int \frac{e^x}{e^{2x} + 1} dx$
 $\text{Let } u = e^x$
 $\frac{du}{dx} = e^x$
 $du = e^x dx$
 $= \int \frac{du}{u^2 + 1}$
 $= \tan^{-1} u + C$
 $= \tan^{-1} e^x + C$



(iv) $y = \sin^{-1}(\sin x)$



(2) $\int \frac{3}{\sqrt{x^2 + 6x + 13}} dx$
 $= \int \frac{3}{\sqrt{x^2 + 6x + 9 - 4 + 13}} dx$
 $= \int \frac{3}{\sqrt{(x+3)^2 + 4}} dx$

$x^2 + 6x + 13$

$= 3 \ln \left| x+3 + \sqrt{(x+3)^2 + 4} \right| + C$

(d) (i) $\int_0^{\pi/2} \frac{1}{1 + \cos x + \sin x} dx = \int_0^1 \frac{1}{1 + \frac{1-t^2}{1+t^2} + \frac{2t}{1+t^2}} \cdot \frac{2dt}{1+t^2}$
 $\left[\text{where } t = \tan \frac{x}{2} \right]$
 $\frac{dx}{dt} = \frac{2}{1+t^2}$

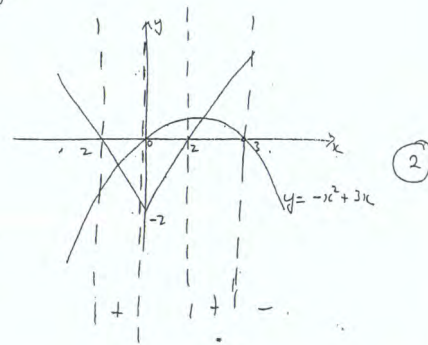
$= \int_0^1 \frac{1+t^2}{1+t^2+1-t^2+2t} \times \frac{2dt}{1+t^2}$
 $= \int_0^1 \frac{1+t^2}{2(1+t)} \times \frac{2dt}{1+t^2}$
 $= \int_0^1 \frac{dt}{1+t}$
 $= \left[\ln|1+t| \right]_0^1$
 $= \ln 2$

(ii) $I = \int_0^{\pi/2} \frac{x}{1 + \cos x + \sin x} dx$
 $\text{Let } u = \frac{\pi}{2} - x$
 $\frac{du}{dx} = -1$

$I = \int_{\pi/2}^0 \frac{\frac{\pi}{2} - u}{1 + \cos(\frac{\pi}{2} - u) + \sin(\frac{\pi}{2} - u)} (-1) du$
 $= \int_0^{\pi/2} \frac{\frac{\pi}{2} - u}{1 + \sin u + \cos u} du$
 $= \int_0^{\pi/2} \frac{\frac{\pi}{2} du}{1 + \sin u + \cos u} - \int_0^{\pi/2} \frac{u du}{1 + \sin u + \cos u}$
 $= \frac{\pi}{2} \int_0^{\pi/2} \frac{du}{1 + \sin u + \cos u}$
 $= \frac{\pi}{2} \times \ln 2$, from (i)

$I = \frac{\pi}{2} \ln 2$

b) (i)



(ii) $-2 < x < 0, 2 < x < 3$

(c) $x^3 + 6x^2 + 12x + 4 = 0$... (1)
 (i) Equat with roots $\frac{1}{x}, \frac{1}{x}, \frac{1}{x}$ will be:
 $\left(\frac{1}{x}\right)^3 + 6\left(\frac{1}{x}\right)^2 + 12\left(\frac{1}{x}\right) + 4 = 0$
 $\times \text{ by } x^3$
 $1 + 6x + 12x^2 + 4x^3 = 0$
 $\therefore 4x^3 + 12x^2 + 6x + 1 = 0$

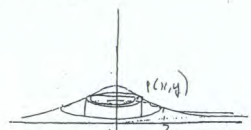
(ii) If $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in A.P.
 $\text{Let } \frac{1}{a} = \frac{1}{b} - d, \frac{1}{c} = \frac{1}{b} + d$
 $\therefore \frac{1}{a} + \frac{1}{c} = \frac{2}{b}$
 $\therefore \frac{1}{a} + \frac{1}{c} = -\frac{12}{4} = -3$
 $\therefore \frac{1}{b} = -1$

Sub. for $x = -1$ in (1)

$\therefore -1 + 6 - 12 + 4 = 0$
 $\therefore b = 9$

80

p6



$$\begin{aligned}
 A &= \pi (R^2 - r^2) \\
 &= \pi (R - r)(R + r) \\
 &= \pi \int_{-r}^r 2x \, dx \\
 &= \pi \int_{-r}^r 2xy \, dx \\
 &= \pi \int_{-r}^r xy \, dx \\
 &= 2\pi \int_0^r x \cdot e^{-\frac{1}{2}x^2} \, dx \\
 &= 2\pi \int_0^r e^{-\frac{1}{2}u} \, du \\
 &= 2\pi \left[\frac{e^{-\frac{1}{2}u}}{-\frac{1}{2}} \right]_0^r \\
 &= -2\pi [e^{-\frac{1}{2}r^2} - 1] \\
 &= 2\pi (1 - e^{-\frac{1}{2}r^2}) \text{ units}^2
 \end{aligned}$$

let $u = x^2$
 $\frac{du}{dx} = 2x$
 $\therefore x dx = \frac{du}{2}$

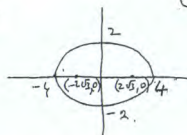
(3)

$$\begin{aligned}
 (p) \quad 19 > l & \quad \text{and } 7 - l > 0 \\
 19 > l & \quad \text{and } 7 - l < 0 \\
 19 > l & \quad l > 7 \\
 7 < l < 19
 \end{aligned}$$

$$(ii) \quad \frac{x^2}{16} + \frac{y^2}{4} = 1$$

$$\begin{aligned}
 b^2 &= a^2 (1 - e^2) \\
 4 &= 16 (1 - e^2) \\
 \frac{1}{4} &= 1 - e^2 \\
 e^2 &= \frac{3}{4} \\
 e &= \frac{\sqrt{3}}{2}
 \end{aligned}$$

$$\text{foci } x = \pm 2\sqrt{3}$$



(2)

$$\begin{aligned}
 (i) \quad \sin(A+B) - \sin(A-B) &= \sin A \cos B + \cos A \sin B - \sin A \cos B + \cos A \sin B \\
 &= 2 \cos A \sin B
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad (1 + 2 \cos \theta + 2 \cos 2\theta + 2 \cos 3\theta) \sin \frac{\theta}{2} &= \sin \frac{\theta}{2} + 2 \cos \theta \sin \frac{\theta}{2} + 2 \cos 2\theta \sin \frac{\theta}{2} + 2 \cos 3\theta \sin \frac{\theta}{2} \\
 &= \sin \frac{\theta}{2} + \sin \frac{3\theta}{2} - \sin \frac{\theta}{2} + \sin \frac{5\theta}{2} - \sin \frac{3\theta}{2} \\
 &= \sin \frac{5\theta}{2}
 \end{aligned}$$

$$\begin{aligned}
 (iii) \quad \text{From (ii)} \quad A \sin \frac{\theta}{2} &= \sin \frac{5\theta}{2} \\
 \text{if } \theta &= \frac{2\pi}{5} \\
 A \sin \frac{2\pi}{5} &= \sin \frac{2\pi}{5} \\
 A \sin \frac{\pi}{5} &= \sin \frac{\pi}{5} = 0 \\
 \Rightarrow A &= 0 \text{ since } \sin \frac{\pi}{5} \neq 0
 \end{aligned}$$

$$\begin{aligned}
 (iv) \quad A &= 1 + 2 \cos \theta + 2 \cos 2\theta + 2 \cos 3\theta \\
 &= 1 + 2 \cos \theta + 2 (2 \cos^2 \theta - 1) \\
 &\quad + 2 (4 \cos^3 \theta - 3 \cos \theta) \\
 &= 1 + 2 \cos \theta + 4 \cos^2 \theta - 2 + 8 \cos^3 \theta - 6 \cos \theta \\
 &= 8 \cos^3 \theta + 4 \cos^2 \theta - 4 \cos \theta - 1
 \end{aligned}$$

$$\begin{aligned}
 (v) \quad \text{If } \theta &= \frac{4\pi}{5}, A = 0 \\
 \text{let } x &= \cos \theta \text{ in (i)} \\
 \therefore A &= 8x^3 + 4x^2 - 4x - 1 = 0 \text{ when } \theta = \frac{4\pi}{5} \\
 \text{is } \cos \frac{4\pi}{5} &\text{ a root of the given cubic}
 \end{aligned}$$

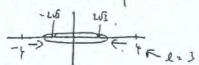
51

(iii) As l increases from 3 to 7,

$$\begin{aligned}
 a^2 &\rightarrow 19 - 7 = 12 \\
 b^2 &\rightarrow 7 - 7 = 0
 \end{aligned}$$

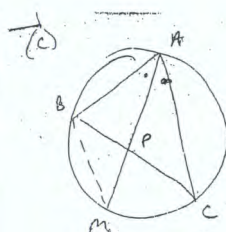
$$a \rightarrow 2\sqrt{3}, b \rightarrow 0$$

so the curve became platter and narrower.

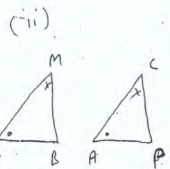


(iv) A straight line on the x axis between $x = -2\sqrt{3}$ and $x = +2\sqrt{3}$, i.e. between the original foci.

(1)



(i) In Δ s ABM & ABC
 $\hat{BAM} = \hat{CAP}$ (given)
 $\hat{BMA} = \hat{BCA}$ (Lr subtends at the circumference by the same arc)
 $\therefore \Delta ABM \sim \Delta ABC$ (equiangular)



$$\begin{aligned}
 \frac{AB}{AP} &= \frac{AM}{AC} \text{ (corresponding sides in } \sim \Delta \text{)} \\
 \therefore AB, AC &= AP, AM
 \end{aligned}$$

$$\begin{aligned}
 (iii) \quad BP, PC &= AP, PM \text{ (intersecting chords in same ratio)} \\
 AB, AC &= BP, PC = AP, PM = AP, PM, \text{ from (ii)} \\
 &= AP (AM - PM) \\
 &= AP \times AP \\
 &= AP^2
 \end{aligned}$$

(3)

$$\begin{aligned}
 (b) \quad (i) \quad I_n &= \int_1^e x (\ln x)^n \, dx \\
 \text{let } u &= (\ln x)^n, v' = x \Rightarrow v = \frac{1}{2}x^2 \\
 I_n &= \left[\frac{1}{2}x^2 (\ln x)^n \right]_1^e - \int_1^e \frac{1}{2}x^2 \cdot n (\ln x)^{n-1} \times \frac{1}{x} \, dx \\
 &= \frac{1}{2}e^2 - \frac{n}{2} \int_1^e x (\ln x)^{n-1} \, dx \\
 &= \frac{1}{2}e^2 - \frac{n}{2} I_{n-1}
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad V &= \pi \int_1^e x (\ln x)^n \, dx \\
 &= \pi I_n = \left(\frac{1}{2}e^2 - \frac{n}{2} I_{n-1} \right) \pi \\
 &\quad + \left(\frac{1}{2}e^2 - \left(\frac{1}{2}e^2 - \frac{1}{2} I_0 \right) \right) \pi \\
 I_0 &= \int_1^e x \, dx \\
 &= \left[\frac{1}{2}x^2 \right]_1^e = \frac{1}{2}(e^2 - 1) \\
 \therefore V &= \pi \left[\frac{1}{2}e^2 - \frac{1}{2}e^2 + \frac{1}{2} \times \frac{1}{2}(e^2 - 1) \right] \\
 &= \frac{\pi}{4} [e^2 - 1] \text{ units}^3
 \end{aligned}$$

$$(c) \quad 20 = 4M, 4A, 6E, 6H$$

$$(i) \quad {}^{20}C_9 = 167960$$

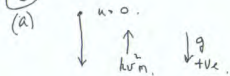
$$(ii) \quad {}^{12}C_9 = 220$$

(iii)	no meth	no A	${}^{12}C_9$	220
	no meth	1 A	${}^4C_1 \times {}^{11}C_8$	1980
	1 meth	no A	${}^4C_1 \times {}^{11}C_8$	1980
	1 meth	1 A	${}^4C_1 \times {}^4C_1 \times {}^{10}C_7$	7920
				12100

$$\begin{aligned}
 \text{Prob} &= \frac{12100}{167960} \\
 &= \frac{605}{8398}
 \end{aligned}$$

(2)

(6)



(i) $m \frac{dv}{dt} = mg - kv^2$ ①

(ii) Terminal vel approached when $\frac{dv}{dt} = 0$ ①

$0 = g - kv^2$
 $v = \sqrt{\frac{g}{k}}$ (+ve $\therefore$ particle going down)

(iii) $v \frac{dv}{dx} = g - kv^2$
 $\frac{dv}{dx} = \frac{g}{v} - kv = \frac{g - kv^2}{v}$ ④

$\frac{dx}{dv} = \frac{v}{g - kv^2}$
 $x = \int \frac{v}{g - kv^2} dv$ let $v^2 = u$
 $\frac{dx}{du} = \frac{1}{2v}$

$= \frac{1}{2} \int \frac{du}{g - ku}$
 $= \frac{1}{2k} \ln(g - kv^2) + C$
 $= \frac{1}{2k} \ln(g - kv^2) + C$

when $x=0, v=0$
 $0 = -\frac{1}{2k} \ln g + C$
 $C = \frac{1}{2k} \ln g$

$x = -\frac{2k}{g} \ln(g - kv^2) + C$

$= \frac{1}{2k} \ln\left(\frac{g}{g - kv^2}\right)$

(iv) when $v = W$
 $S = \frac{1}{2k} \ln\left(\frac{g}{g - kW^2}\right)$

From (ii), $v = \sqrt{\frac{g}{k}}$
 $\therefore g = kv^2$

$S = \frac{1}{2k} \ln\left(\frac{kv^2}{kv^2 - kW^2}\right)$
 $= \frac{1}{2k} \ln\left(\frac{v^2}{v^2 - W^2}\right)$

①

r_1, r_2, r_3, r_4 are roots of the equation coming from sub

$x = ct + y = \frac{c}{t}$ in
 $(x-g)^2 + y^2 = r^2$

$(ct-g)^2 + \left(\frac{c}{t}\right)^2 = r^2$
 $c^2 t^2 - 2gct + g^2 + \frac{c^2}{t^2} = r^2$ ②

$c^2 t^4 - 2gct^3 + g^2 t^2 + c^2 = r^2 t^2$
 $c^2 t^4 - 2gct^3 + g^2 t^2 - r^2 t^2 + c^2 = 0$

$t_1 + t_2 + t_3 + t_4 = \frac{2gc}{c^2} = \frac{2g}{c}$

(ii) $C = (g, 0)$
 $\therefore g = \frac{ct_1 + ct_2}{2}$
 $\therefore t_1 + t_2 = \frac{2g}{c}$ ②

(iii) Midpt of $AB = M = \left[\frac{c(t_1+t_2)}{2}, \frac{1}{2}\left(\frac{c}{t_1} + \frac{c}{t_2}\right) \right]$
 $= (x_1, x_2)$ ②

$x_1 = \frac{c}{2} \left(\frac{2g}{c} - t_1 - t_2 \right)$
 $= \frac{c}{2} \left(\frac{2g}{c} - \frac{2g}{c} \right) = 0$
 $y_1 = \frac{c}{2} \left(\frac{t_1+t_2}{t_1 t_2} \right)$
 $= 0$, since $t_1 t_2 = \frac{2g}{c} - (t_1 + t_2) = 0$ from (ii)

$15 \sin \alpha + 6 \sin \alpha$ ①

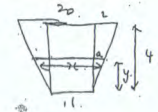
(ii) $x^2 = 36 + 25 - 2 \times 70 \cos \beta = 9 + 16 - 2 \times 12 \cos \beta$
 $24 \cos \alpha = 60 \cos \beta = -36$
 $\therefore 2 \cos \alpha = 5 \cos \beta = -2$ ①

(iii) $-2 \sin \alpha + 5 \sin \beta \cdot \frac{d\beta}{d\alpha} = 0$
 $\frac{d\beta}{d\alpha} = \frac{2 \sin \alpha}{5 \sin \beta}$ ①

(iv) $\frac{dA}{d\alpha} = 6 \cos \alpha + 15 \cos \beta \cdot \frac{d\beta}{d\alpha} = 0$ for a max
 $\therefore 0 = 6 \cos \alpha + 15 \cos \beta \cdot \frac{2 \sin \alpha}{5 \sin \beta}$
 $0 = \frac{6}{\sin \beta} (\sin \beta \cos \alpha + \sin \alpha \cos \beta)$ ③

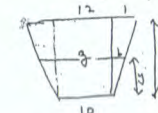
$0 = \sin(\alpha + \beta)$
 $\Rightarrow \alpha + \beta = 0, \pi, 2\pi \rightarrow 0$ impossible since α, β are impossible since sum of 4 hrs = 2
 $\therefore \alpha + \beta = \pi$
 $\therefore$ the quad is cyclic (opp hrs supp)

b) (i)



$\frac{2}{y} = \frac{8}{y}$
 $x - 2a = 16$
 $a = \frac{x-16}{2}$
 $\frac{1}{2} = \frac{x-16}{y}$
 $y = x - 16$
 $x = (y+16)$

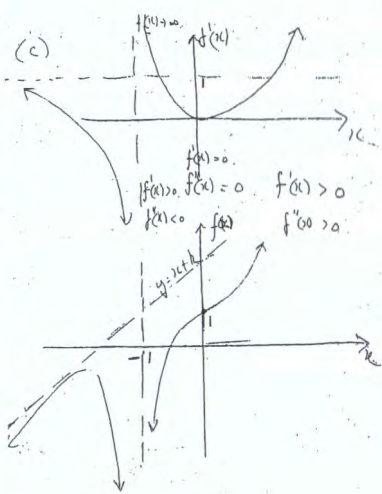
③



$\frac{1}{y} = \frac{1}{y}$
 $4b = y$
 $2g - 10 = y$
 $g = \frac{y}{2} + 10$

Area = $(y+16)\left(\frac{y}{2} + 10\right)$
 $= \frac{1}{2} y^2 + 18y + 160$ ②

(ii) $Vol = \lim_{dy \rightarrow 0} \sum_{y=0}^y \frac{1}{2} y^2 + 18y + 160 \cdot dy$
 $= \int_0^y \left(\frac{1}{2} y^2 + 18y + 160 \right) dy$
 $= \left[\frac{1}{6} y^3 + 9y^2 + 160y \right]_0^y$
 $= \frac{1}{6} \times 64 + 9 \times 16 + 160 \times 4$
 $= 794 \frac{2}{3}$ m³



p12

③

(v) $2 \cos \alpha - 5 \cos \beta = -3$
 $\therefore 2 \cos \alpha - 5 \cos(\pi - \alpha) = -3$
 $2 \cos \alpha + 5 \cos \alpha = -3$
 $7 \cos \alpha = -3$
 $\cos \alpha = -\frac{3}{7}$ ✓



$A = 6 \sin \alpha + 15 \sin(\pi - \alpha)$
 $= 6 \times \frac{\sqrt{40}}{7} + 15 \times \frac{\sqrt{40}}{7}$
 $= \left(\frac{12}{7} + \frac{45}{7} \right) \sqrt{10}$
 $= 6 \sqrt{10}$ units² ②

⑧ (a) (i) $\cos A = \cos B$
 $\therefore A = 2n\pi \pm \cos^{-1}(\cos B)$
 $= 2n\pi \pm B, n$ is an integer. ①

(ii) $\cos 2\theta = \sin 3\theta, 0 \leq \theta \leq \pi$
 $= \cos\left(\frac{\pi}{2} - 3\theta\right)$
 $\therefore 2\theta = 2n\pi \pm \left(\frac{\pi}{2} - 3\theta\right)$ ③

$2\theta = 2n\pi + \frac{\pi}{2} - 3\theta$ or $2\theta = 2n\pi - \frac{\pi}{2} + 3\theta$
 $5\theta = 2n\pi + \frac{\pi}{2}$ or $\theta = \frac{2n\pi + \frac{\pi}{2}}{5}$
 $n=0 \Rightarrow \theta = \frac{\pi}{10}$
 $n=1 \Rightarrow \theta = \frac{2\pi + \frac{\pi}{2}}{5} = \frac{5\pi + \frac{\pi}{2}}{10} = \frac{11\pi}{20}$
 $n=2 \Rightarrow \theta = \frac{4\pi + \frac{\pi}{2}}{5} = \frac{9\pi}{10}$
 $\theta = \frac{\pi}{10}, \frac{11\pi}{20}, \frac{9\pi}{10}$

Final solutions: $\theta = \frac{\pi}{10}, \frac{11\pi}{20}, \frac{9\pi}{10}$

p14

(b) (i) Show $\sum_{r=1}^n \frac{1}{\sqrt{r}} > 2(\sqrt{n+1} - 1)$

Let $n=1$: LHS = 1 RHS = $2(\sqrt{2} - 1)$
 $= 0.828$

True for $n=1$

assume true for $n=k$

i.e. $\sum_{r=1}^k \frac{1}{\sqrt{r}} > 2(\sqrt{k+1} - 1)$ — (1)

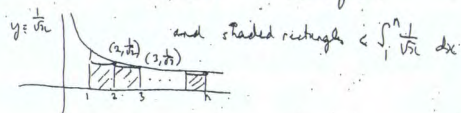
now true for $n=k+1$

i.e. $\sum_{r=1}^{k+1} \frac{1}{\sqrt{r}} > 2(\sqrt{k+2} - 1)$

proof: $S_{k+1} = S_k + \frac{1}{\sqrt{k+1}}$
 $> 2(\sqrt{k+1} - 1) + \frac{1}{\sqrt{k+1}}$ from (1)
 $= \frac{2(k+1) - 2\sqrt{k+1} + 1}{\sqrt{k+1}}$
 $= \frac{2k+3 - 2\sqrt{k+1}}{\sqrt{k+1}}$
 $> \frac{2\sqrt{(k+1)(k+1)} - 2\sqrt{k+1}}{\sqrt{k+1}}$, given property
 $= 2(\sqrt{k+2} - 1)$

∴ true for $n=1, 2, \dots, k, k+1$, ∴ True $\forall$ integers n by induction.

(ii) $\sum_{r=1}^n \frac{1}{\sqrt{r}} = 1 + \frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{n}}$
 $= 1 + \text{shaded rectangles}$

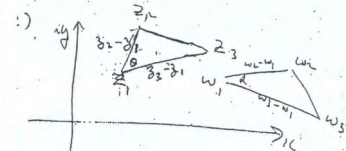


$\sum_{r=1}^n \frac{1}{\sqrt{r}} < 1 + \int_1^n \frac{1}{\sqrt{x}} dx$

(iii) $\sum_{r=1}^{10000} \frac{1}{\sqrt{r}} < 1 + \int_1^{10000} \frac{1}{\sqrt{x}} dx$
 $= 1 + [2\sqrt{x}]_1^{10000}$
 $= 1 + 2 \times 100 - 2$
 $= 199$

also $\sum_{r=1}^{10000} \frac{1}{\sqrt{r}} > 2(\sqrt{10000+1} - 1)$ from (i)
 $= 198.00\dots$

$\therefore 198 < \sum_{r=1}^{10000} \frac{1}{\sqrt{r}} < 199$



Let $\frac{z_2 - z_1}{z_3 - z_1} = \frac{w_2 - w_1}{w_3 - w_1}$, let $\theta + 2\pi$ be $\angle$ shown

then $\left| \frac{z_2 - z_1}{z_3 - z_1} \right| = \left| \frac{w_2 - w_1}{w_3 - w_1} \right|$

$\frac{|z_2 - z_1|}{|z_3 - z_1|} = \frac{|w_2 - w_1|}{|w_3 - w_1|}$, i.e. corresponding sides in same ratio

also $\arg \left(\frac{z_2 - z_1}{z_3 - z_1} \right) = \arg \left(\frac{w_2 - w_1}{w_3 - w_1} \right)$, since $\frac{z_2 - z_1}{z_3 - z_1} = \frac{w_2 - w_1}{w_3 - w_1}$

i.e. $\arg(z_2 - z_1) - \arg(z_3 - z_1)$
 $= \arg(w_2 - w_1) - \arg(w_3 - w_1)$

i.e. $\theta = \alpha$

$\Delta z_1 z_2 z_3 \parallel \Delta w_1 w_2 w_3$

(SAS similarity)